Artificial Intelligence and Data Science

Data Science and Visualization

24ADI204

MID REVIEW (September 2026)

Team Number: 3

DATA-VIZ HACK-A-BIT (2026-2027)

NETFLIX CONTENTS PATTERNS ANALYSIS

Team Members

Dakshata SP – Data Science Engineer

Rakshita A. – Data Analyst

Sahana Viyani A. – Data Visualizer

Problem Statement

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- Netflix has a large collection of Movies and TV Shows.
- It is difficult to understand the content distribution manually.
- We need to identify popular content types, ratings, genres and release-year patterns.
- We also want to understand content characteristics related to binge-watching potential.
- Data analysis and visualization can help us find these patterns easily.

Simple version:

To analyze Netflix content data and identify important patterns in genres, ratings, content types, duration and TV-show seasons.

Project Objectives

Main Objectives

- **Understand the Netflix dataset structure.**
- **Identify missing values and data-quality issues.**
- **Clean and prepare the dataset for analysis.**
- **Analyze Movies and TV Shows.**
- **Study ratings, genres, release years and duration.**
- **Identify patterns related to binge-watching potential.**
- **Present findings using charts and visualizations.**

Week 2 – Know Your Data

- Load and understand the Netflix dataset.
- Check rows, columns and data types.
- Identify missing values and duplicates.
- Generate first insights from the raw data.

Week 3 – Data Cleaning

- Handle missing values.
- Clean and standardize data.
- Separate movie duration and TV-show seasons.
- Detect unusual values using IQR and Z-score.

Week 4 – EDA Deep Dive

- Analyze individual variables.
- Compare two variables using bivariate analysis.
- Study genres, ratings, release years and duration.
- Find important patterns using graphs and statistics.

Dataset Description

Dataset: Netflix Movies and TV Shows

Source: Kaggle – Netflix Movies and TV Shows Dataset

Dataset Size:

- 8,807 records
- 12 columns

Important Columns:

- type – Movie / TV Show
- title – Content name
- director – Director
- cast – Actors
- country – Country
- release_year – Release year
- rating – Content rating
- duration – Movie duration / TV seasons
- listed_in – Genre/category
- description – Content description

Methodology(Process / Workflow)

Project Workflow

Dataset Collection

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Data Loading

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Data Understanding

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Data Cleaning

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Feature Extraction

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Outlier Detection

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Exploratory Data Analysis

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Visualization

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Insights & Conclusion

Week Mapping

- Week 2: Understand Data
- Week 3: Clean Data
- Week 4: Analyze Data

Results From Workflow

Data Quality Improvements

- Missing values were identified and treated.
- Duration was separated into meaningful numerical features.
- Duplicate records were checked.
- Outliers were identified for further understanding.

Key EDA Findings

- Movies: 6,131
- TV Shows: 2,676
- Top category: International Movies
- Movie duration is centred around 98 minutes.
- TV Shows commonly have 1 season.

Dataset Readiness Status

- Dataset cleaned and prepared for further analysis.
- EDA completed for Weeks 2–4.
- Dataset is ready for the next stages of the project.

Expected Final Results

- Identify major Netflix content patterns.
- Understand popular genres and ratings.
- Analyze movie duration and TV-show season patterns.
- Identify catalogue-level indicators related to binge-watching.
- Connect findings with SDG 3 and SDG 4.

Roles & Responsibilities

Team Member	Roles & Responsibilities
Dakshata SP	Data Engineer
Rakshita . A	Data Analyst
Sahana Viyani . A	Data Visualizer

Future Work & Conclusion

- Use viewer-level data to study actual binge-watching behaviour.
- Analyze watch time, binge sessions and episode completion.
- Develop further visualizations and dashboards.
- Current analysis provides a clear understanding of the Netflix catalogue.

Challenges Faced

- Many missing values in descriptive fields.
- Duration column had different units for Movies and TV Shows.
- Some unusual values required data-quality checking.
- The dataset does not contain individual viewer watch history, so actual binge-watching behaviour cannot be directly measured.

References

- Kaggle – Netflix Movies and TV Shows Dataset
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Project Google colab,VS code